

such as the arts, the odds of success are low, since there are fewer successful people, but the payoff is disproportionately high. So, in a fat-tailed environment, rare events can be less frequent (their probability is lower), but they are so powerful that their contribution to the total pie is more substantial.

The point is mathematically simple, but does not register easily. I've enjoyed giving graduate students in mathematics the following quiz (to be answered intuitively, on the spot). In a Gaussian world, the probability of exceeding one standard deviation is around 16 percent. What are the odds of exceeding it under a distribution of fatter tails (with the same mean and variance)? The right answer: lower, not higher—the number of deviations drops, but the few that take place matter more. It was puzzling to see that most graduate students get it wrong.

Back to stress testing again. At the time of writing, the U.S. government is having financial institutions stress-tested by assuming large deviations and checking the results against the capitalization of these firms. But the problem is, Where did they get the numbers? From the past? This is so flawed, since the past, as we saw, is no indication of future deviations in Extremistan. This comes from the atypicality of extreme deviations. My experience of stress testing is that it reveals little about the risks—but the risks can be used to assess the degree of model error.

Psychology of Perception of Deviations

Fragility of Intuitions About the Typicality of the Move. Dan Goldstein and I ran a series of experiments about the intuitions of agents concerning such conditional expectations. We posed questions of the following sort: What is the average height of humans who are taller than six feet? What the average weight of people heavier than 250 pounds? We tried with a collection of variables from Mediocristan, including the above-mentioned height and weight, to which we added age, and we asked participants to guess variables from Extremistan, such as market capitalization (what is the average size of companies with capitalization in excess of \$5 billion?) and stock performance. The results show that, clearly, we have good intuitions when it comes to Mediocristan, but horribly poor ones when it comes to Extremistan—yet economic life is almost all Extremistan. We do not have good intuition for that atypicality of large deviations. This explains both foolish risk taking and how people can underestimate opportunities.

Framing the Risks. Mathematically equivalent statements, I showed